

Laurent Colbois

ML ENGINEER · PHD IN MACHINE LEARNING & BIOMETRICS

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“To create a positive impact through technology while following sustainable practices.”

Summary

Machine Learning Engineer with PhD from UNIL/Idiap and Master's in Computational Science & Engineering from EPFL, combining expertise in deep learning with foundations in numerical simulations and physics-based modeling. Specialized in bridging ML and engineering domains through reproducible workflows and clear technical communication. Proven ability to translate complex concepts across audiences—from forensic experts to investors. Eager to apply ML to real-world engineering challenges at the intersection of physics, simulation, and AI.

Work Experience

Idiap Research Institute

Martigny, Switzerland

POSTDOCTORAL RESEARCHER - VISION-LANGUAGE MODELS FOR BIOMETRICS

Apr. 2025 - Mar. 2026

- Benchmarked open-weight Vision-Language Models (Gemma3, Qwen2.5-VL, InternVL3) for explainable face recognition, developing novel metrics to quantify explanation quality and occlusion-faithfulness
- Built end-to-end reproducible ML pipelines using Hydra for configuration, Weights & Biases for tracking, and vLLM for efficient inference—emphasizing code quality and workflow automation
- Led cross-functional project coordination for a forensic data collection platform, managing communication between researchers, developers, and legal staff while defining technical specifications and prioritizing requirements
- Won first place (15K CHF) at 9-day startup hackathon by developing real-time MIDI music generation demo and successfully pitching AI-based music improvisation partner to technical and business audiences

Idiap Research Institute

Martigny, Switzerland

PHD STUDENT - DEEPFAKES, BIOMETRICS, FORENSIC SCIENCE

Jan. 2021 - Mar. 2025

- Developed detection systems for AI-generated images with focus on fake passport photos, including vulnerability assessment of face recognition systems to deepfakes
- Presented technical concepts on AI and deepfakes to non-technical stakeholders including forensic science experts, police forces, and general public—adapting messaging to diverse audience backgrounds
- Supervised four master's students on score calibration projects for forensic face recognition, providing technical mentorship and project guidance
- Published reproducible research code packages emphasizing experiments reproducibility and workflow automation

Idiap Research Institute

Martigny, Switzerland

RESEARCH INTERN - SYNTHETIC DATA FOR FACE RECOGNITION

Jul. - Dec. 2020

- Created and published synthetic face recognition dataset using generative AI (GANs) and evaluated its applicability for benchmarking face recognition systems

Logitech

Lausanne, Switzerland

RESEARCH INTERN - AUDIO MACHINE LEARNING

Feb. - Aug. 2019

- Developed automated audio scene classification model for auto-equalization of loudspeaker and implemented real-time demo on embedded microcontroller

Education

University of Lausanne / Idiap Research Institute

Lausanne, Switzerland

PHD IN FORENSIC SCIENCE (MACHINE LEARNING & BIOMETRICS)

2021 - 2025

- Thesis: Face morphing attacks in the era of deepfakes: risks, detection & source attribution

- Thesis: Trainable filterbanks for end-to-end speech enhancement via time-frequency masking
- Skills: Numerical simulations, Machine learning, Deep learning, Signal processing, High performance computing

- Skills: Applied mathematics, Numerical physics, Programming, Scientific computing

1 Publications

2026	Conference Paper (Under Review) , Benchmarking the Explanatory Quality of Open-Weight Vision-Language Models in Face Recognition	<i>Face and Gesture</i>
2024	Conference Paper , Evaluating the effectiveness of attack-agnostic features for morphing attack detection	<i>IJCB</i>
2023	Conference Paper , Approximating Optimal Morphing Attacks using Template Inversion	<i>IJCB</i>
2022	Conference Paper , On the detection of morphing attacks generated by GANs	<i>BIOSIG</i>
2022	Conference Paper , Are GAN-based morphs threatening face recognition?	<i>ICASSP</i>
2021	Conference Paper , On the use of automatically generated synthetic image datasets for benchmarking face recognition	<i>IJCB</i>

Skills

Domains	Machine Learning for engineering & physics, Computer vision, Deep learning, Vision-Language Models, Numerical simulations (discretization, solvers), Reproducible research, Cross-functional communication
ML & Programming	Python, PyTorch, Huggingface Transformers, vLLM, Scikit-Learn, Pandas, NumPy, C++, MATLAB
MLOps & DevOps	Git, Docker, Apptainer/Singularity, Conda, Pip, Pixi, Snakemake, Prefect, Weights & Biases, MLFlow, Hydra, Slurm, Unix/Linux
Languages	French (Mother tongue), English (Professional proficiency), German (B2)